

MATH 2123 (MATHEMATICS-III)

2015 Beginner Guide + Full Step-by-Step Solutions

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NOTE: This guide follows the two supplied pages of the 2015 paper. It is written for a student who is not yet confident in the topics. Each set is arranged as Topic -> Basic -> Problem-Solving Procedure -> Worked Solution. In Question 6(c), r^2 is interpreted in the standard vector-calculus sense as $x^2+y^2+z^2$.

How to study from this PDF

- Read the BASIC box before the solution. It contains the minimum idea you need.
- Learn the PROCEDURE as a reusable recipe. The numbers change in exams; the procedure usually does not.
- Then follow the solution line by line and check the green final-answer box.
- Use the Rapid Revision section at the end for last-minute recall.

Question map

Set	Main topic	Core skill
1	Matrix basics + vector dependence	definitions, matrix arithmetic exceptions, dependence
2	Rank, invertibility, inverse	row reduction, normal form, elementary matrices
3	Parametric systems + Markov eigenvalues	determinant cases, dominant eigenpair
4	Eigenvalue properties + Cayley- Hamilton + homogeneous systems	characteristic equation, nullspace
5	Space-curve kinematics + gradient/divergence	differentiate/integrate vectors
6	Surfaces, conservative fields, line integrals	gradients, potential, curve parameterization
7	Surface flux + Laplace properties	surface parameterization, transform rules
8	Laplace and inverse Laplace	differentiation in s, shifts, logs, convolution

Question 1: Matrix basics and linear dependence

TOPIC	BASIC	PROCEDURE
Matrices and vector dependence	Equivalent matrices have the same size and corresponding entries. A singular square matrix has $\det=0$. $\text{trace}(A)$ is the sum of diagonal entries. Symmetric means $A^T=A$. Vectors are dependent if a nonzero coefficient combination gives zero.	Definitions: state condition + simple example. Matrix-arithmetic exceptions: show explicit counterexamples. Dependence: solve $c_1v_1+c_2v_2+c_3v_3=0$; a nonzero solution proves dependence.

1(a) - Definitions with examples

The paper asks for any three; all four are included for revision.

Equivalent matrices: two matrices are equal/equivalent in the elementary sense here when they have the same order and equal corresponding entries.

Example: $[[1,2],[3,4]] = [[1,2],[3,4]]$.

Singular matrix: a square matrix A is singular if $\det(A)=0$, so A^{-1} does not exist.

Example: $[[1,2],[2,4]]$, $\det = 4-4 = 0$.

Trace: $\text{tr}(A)$ is the sum of the principal diagonal entries.

Example: $\text{tr}([[2,1],[5,3]]) = 2+3 = 5$.

Symmetric matrix: $A^T=A$.

Example: $[[2,4],[4,7]]$ is symmetric.

FINAL ANSWER: Definitions and examples are as listed above; in the exam, write any three clearly.

1(b) - Three ways matrix arithmetic differs from real-number arithmetic

Case 1 - multiplication is generally not commutative.

$A=[[1,1],[0,1]]$, $B=[[1,0],[1,1]]$

$AB=[[2,1],[1,1]]$, but $BA=[[1,1],[1,2]]$. Hence $AB \neq BA$.

Case 2 - a zero product does not force one factor to be zero.

$A=[[1,0],[0,0]]$, $B=[[0,0],[0,1]]$. $A \neq 0$, $B \neq 0$, but $AB=0$.

Case 3 - cancellation can fail.

$A=[[1,0],[0,0]]$, $B=[[1,0],[0,0]]$, $C=[[1,0],[0,1]]$.

$AB=AC=[[1,0],[0,0]]$, but $B \neq C$.

FINAL ANSWER: Matrices need not commute; nonzero matrices can multiply to zero; cancellation can fail.

1(c) - Test the three $\mathbb{R}^4$ vectors for linear dependence

$v_1=(0,1,2,3)$, $v_2=(2,8,9,1)$, $v_3=(2,6,5,-5)$

Assume $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$. Component equations are

$$2c_2+2c_3=0$$

$$c_1+8c_2+6c_3=0$$

$$2c_1+9c_2+5c_3=0$$

$$3c_1+c_2-5c_3=0$$

A nonzero solution is $(c_1, c_2, c_3) = (2, -1, 1)$. Therefore

$$2v_1 - v_2 + v_3 = 0$$

Equivalently,

$$v_2 = 2v_1 + v_3 \quad (\text{or } v_3 = v_2 - 2v_1)$$

FINAL ANSWER: The vectors are linearly dependent. One verified relation is $2v_1 - v_2 + v_3 = 0$.

Question 2: Rank, invertibility and elementary matrices

TOPIC	BASIC	PROCEDURE
Row reduction and inverse	Rank is the number of pivots. A square matrix is invertible iff $\det \neq 0$ (equivalently full rank). Elementary row operations correspond to multiplication by elementary matrices.	Echelon: create zeros below pivots. Canonical/RREF: make pivots 1 and clear pivot columns. Normal form: also use column operations to obtain $[I_r \ 0; \ 0 \ 0]$. For inverse, row-reduce $[A I]$ or track elementary matrices.

2(a) - Echelon, canonical and normal forms; rank

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 2 & 1 & 2 & 1 \\ 4 & 2 & 0 & 2 \end{bmatrix}$$

Use the following row operations:

1. $R_1 \leftrightarrow R_2$.
2. $R_3 \rightarrow R_3 - 2R_1$.
3. $R_3 \rightarrow R_3 + 4R_2$.

$$\text{Echelon form} = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now $R_1 \rightarrow (1/2)R_1$ and then $R_1 \rightarrow R_1 - R_2$:

$$\text{Row-canonical (RREF)} = \begin{bmatrix} 1 & 1/2 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

For the normal form, use column operations $C_2 \rightarrow C_2 - (1/2)C_1$, $C_4 \rightarrow C_4 - (1/2)C_1$, then $C_2 \leftrightarrow C_3$.

$$\text{Normal form} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

FINAL ANSWER: $\text{rank}(A) = 2$.

2(b) - Which of the listed matrices are invertible?

$$A_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \det(A_1) = -1 \neq 0, \text{ so invertible.}$$

$$A_2 = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \Rightarrow \det(A_2) = 4 - 4 = 0, \text{ so not invertible.}$$

$$A_3 = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \text{ is } 2 \times 3, \text{ not square, so an ordinary inverse does not exist.}$$

FINAL ANSWER: Only the first matrix is invertible.

2(c) - Find A^{-1} and express A and A^{-1} as products of elementary matrices

$$A = \begin{bmatrix} 0 & 1 \\ 2 & 4 \end{bmatrix}, \det(A) = -2 \neq 0$$

$$A^{-1} = \begin{bmatrix} -2 & 1/2 \\ 1 & 0 \end{bmatrix}$$

Reduce A to I :

1. E_1 : swap R_1 and R_2 .
2. E_2 : $R_1 \rightarrow (1/2)R_1$.
3. E_3 : $R_1 \rightarrow R_1 - 2R_2$.

$$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$E_2 = \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix}$$

$$E_3 = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$E_3 E_2 E_1 A = I \Rightarrow A^{-1} = E_3 E_2 E_1$$

Therefore A is the product of the inverse elementary matrices in reverse order:

$$A = E_1^{-1} E_2^{-1} E_3^{-1}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

FINAL ANSWER: $A^{-1} = \begin{bmatrix} -2 & 1/2 \\ 1 & 0 \end{bmatrix}$, with elementary-matrix products shown above.

Question 3: Parameter-dependent system and Markov eigenvalues

TOPIC

Determinant cases and eigenpairs

BASIC

For a 2×2 system, $\det(\text{coefficient matrix}) \neq 0$ gives a unique solution. When $\det = 0$, compare the equations for consistency. Eigenvalues solve $\det(A - \lambda I) = 0$; A^n has eigenvalues λ^n with the same eigenvectors.

PROCEDURE

System: compute determinant $\rightarrow$ inspect special λ values. Markov matrix: solve characteristic equation $\rightarrow$ choose eigenvector $\rightarrow$ verify $Av = \lambda v \rightarrow$ use power property for A^{10} .

3(a) - Values of λ for unique, many, or no solution

$$\lambda x + y = 2$$

$$x + \lambda y = 2\lambda$$

$$\text{Coefficient determinant} = \lambda^2 - 1$$

If $\lambda \neq \pm 1$, determinant is nonzero, so the solution is unique. Solving gives $x=0, y=2$.

If $\lambda=1$: both equations become $x+y=2$, so there are infinitely many solutions.

If $\lambda=-1$: the equations are $-x+y=2$ and $x-y=-2$; these are the same equation, so again infinitely many solutions.

FINAL ANSWER: Unique: $\lambda \neq \pm 1$. Infinitely many: $\lambda=1$ or $\lambda=-1$. No-solution case: none.

3(b) - Largest eigenvalue/eigenvector of the Markov matrix and of A^{10}

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

$$\begin{aligned}\det(A - \lambda I) &= (0.8 - \lambda)(0.7 - \lambda) - 0.06 \\ &= \lambda^2 - 1.5\lambda + 0.5 \\ &= (\lambda - 1)(\lambda - 0.5)\end{aligned}$$

The largest eigenvalue is $\lambda = 1$.

$$(A - I)v = 0 \Rightarrow -0.2x + 0.3y = 0 \Rightarrow 2x = 3y$$

Choose $x = 3$, $y = 2$:

$$v = (3, 2)^T, \text{ and } Av = (3, 2)^T = 1 \cdot v$$

For A^{10} , the eigenvalues are $1^{10} = 1$ and $(0.5)^{10} = 1/1024$. Eigenvectors are unchanged.

FINAL ANSWER: Largest eigenvalue of A and A^{10} is 1; a corresponding eigenvector is $(3, 2)^T$ (normalized Markov vector: $(3/5, 2/5)^T$).

Question 4: Eigenvalue properties, Cayley-Hamilton and a homogeneous system

TOPIC	BASIC	PROCEDURE
Eigensystem and nullspace	Useful facts: sum of eigenvalues (with multiplicity) = trace; product = determinant; eigenvalues of A^n are λ^n . Cayley-Hamilton: a square matrix satisfies its own characteristic polynomial. A homogeneous system always has the trivial solution; a nontrivial solution exists when the coefficient matrix is singular.	State clean properties. For C-H: form $p(\lambda)$, then set $p(A) = 0$. For a homogeneous system: Gaussian eliminate; take a free variable as a parameter and back-substitute.

4(a) - Three properties of eigenvalues

- The sum of eigenvalues (counting algebraic multiplicity) equals $\text{tr}(A)$.
- The product of eigenvalues equals $\det(A)$.
- If λ is an eigenvalue of A , then λ^n is an eigenvalue of A^n for the same eigenvector.

FINAL ANSWER: Any three correct standard properties earn the requested marks; three are listed above.

4(b) - Use Cayley-Hamilton for A^2 and A^{-2} , if possible

$$A = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$$

$$p(\lambda) = \lambda^2 - \text{tr}(A)\lambda + \det(A) = \lambda^2 - 8\lambda$$

By Cayley-Hamilton,

$$A^2 - 8A = 0 \Rightarrow A^2 = 8A = \begin{bmatrix} 16 & 24 \\ 32 & 48 \end{bmatrix}$$

But $\det(A) = 12 - 12 = 0$. Therefore A is singular and A^{-1} , hence A^{-2} , does not exist.

FINAL ANSWER: $A^2 = \begin{bmatrix} 16 & 24 \\ 32 & 48 \end{bmatrix}$. A^{-2} does not exist.

4(c) - Find the nontrivial solutions of the homogeneous system

$$4x_1 - x_2 + 2x_3 + x_4 = 0$$

$$2x_1 + 3x_2 - x_3 - 2x_4 = 0$$

$$7x_1 - 4x_2 - 5x_4 = 0$$

$$2x_1 - 11x_2 + 7x_3 + 8x_4 = 0$$

Use the second equation as the first pivot row. A convenient elimination gives

$$[2, 3, -1, -2]$$

$$[0, -7, 4, 5]$$

$$[0, 0, -67, -117]$$

$$[0, 0, 0, 0]$$

Let $x_4 = 67k$. Then

$$-67x_3 - 117x_4 = 0 \Rightarrow x_3 = -117k$$

$$-7x_2 + 4(-117k) + 5(67k) = 0 \Rightarrow x_2 = -19k$$

$$2x_1 + 3(-19k) - (-117k) - 2(67k) = 0 \Rightarrow x_1 = 37k$$

FINAL ANSWER: $(x_1, x_2, x_3, x_4) = k(37, -19, -117, 67)$, with $k \neq 0$ for a nontrivial solution.

Question 5: Vector kinematics and physical meaning of grad/div

TOPIC	BASIC	PROCEDURE
Motion in 3D	Position $\mathbf{r}(t) \rightarrow$ velocity $\mathbf{v} = \mathbf{r}'(t) \rightarrow$ acceleration $\mathbf{a} = \mathbf{r}''(t)$. If acceleration is known, integrate twice and use initial velocity/position. grad $\mathbf{f}$ points toward steepest increase; div $\mathbf{F}$ measures local net outflow/source strength.	Differentiate or integrate each component independently. Apply vector initial conditions component by component. For magnitudes use $\sqrt{\mathbf{v} \cdot \mathbf{v}}$ or $\sqrt{\mathbf{a} \cdot \mathbf{a}}$.

5(a) - Velocity and acceleration along the given curve

$$\mathbf{r}(t) = (2\sin 3t, 2\cos 3t, 8t)$$

$$\mathbf{v}(t) = (6\cos 3t, -6\sin 3t, 8)$$

$$\mathbf{a}(t) = (-18\sin 3t, -18\cos 3t, 0)$$

$$|\mathbf{v}| = \sqrt{36\cos^2(3t) + 36\sin^2(3t) + 64} = 10$$

$$|\mathbf{a}| = \sqrt{324\sin^2(3t) + 324\cos^2(3t)} = 18$$

At $t = \pi/6$, $3t = \pi/2$:

$$\mathbf{v} = (0, -6, 8), \mathbf{a} = (-18, 0, 0)$$

NOTE: The paper wording says "along x axis". At this time the x-component of velocity is 0, while the acceleration is entirely along the negative x-axis. The full magnitudes are $|\mathbf{v}| = 10$ and $|\mathbf{a}| = 18$.

FINAL ANSWER: $\mathbf{v} = (6\cos 3t, -6\sin 3t, 8)$, $\mathbf{a} = (-18\sin 3t, -18\cos 3t, 0)$; at $t = \pi/6$: $\mathbf{v} = (0, -6, 8)$, $\mathbf{a} = (-18, 0, 0)$, with magnitudes 10 and 18.

5(b) - Integrate the given vector acceleration twice

$$\mathbf{q}''(t) = 12t \mathbf{i} + 24t^2 \mathbf{j} + \sin(t) \mathbf{k}$$

Integrate once:

$$\mathbf{q}'(t) = (6t^2 + C_1)\mathbf{i} + (8t^3 + C_2)\mathbf{j} + (-\cos t + C_3)\mathbf{k}$$

Given $\mathbf{q}'(0) = 2\mathbf{i} + 3\mathbf{j}$, so $C_1 = 2$, $C_2 = 3$, $C_3 = 1$.

$$\mathbf{q}'(t) = (6t^2 + 2)\mathbf{i} + (8t^3 + 3)\mathbf{j} + (1 - \cos t)\mathbf{k}$$

Integrate again:

$$\mathbf{q}(t) = (2t^3 + 2t + D_1)\mathbf{i} + (2t^4 + 3t + D_2)\mathbf{j} + (t - \sin t + D_3)\mathbf{k}$$

Given $\mathbf{q}(0) = \mathbf{i} + \mathbf{j}$, hence $D_1 = 1$, $D_2 = 1$, $D_3 = 0$.

$$\mathbf{q}(t) = (2t^3 + 2t + 1)\mathbf{i} + (2t^4 + 3t + 1)\mathbf{j} + (t - \sin t)\mathbf{k}$$

$$\mathbf{q}(5) = 261 \mathbf{i} + 1266 \mathbf{j} + (5 - \sin 5) \mathbf{k}$$

FINAL ANSWER: $\mathbf{q}(t) = (2t^3 + 2t + 1)\mathbf{i} + (2t^4 + 3t + 1)\mathbf{j} + (t - \sin t)\mathbf{k}$; $\mathbf{q}(5) = 261\mathbf{i} + 1266\mathbf{j} + (5 - \sin 5)\mathbf{k}$.

5(c) - Physical meaning of grad f and div F

- $\text{grad } f$ (written ∇f) points in the direction of maximum increase of the scalar field f . Its magnitude is the maximum directional derivative.
- $\text{div } F$ (written $\nabla \cdot F$) measures net outward flux per unit volume near a point. Positive divergence behaves like a source; negative divergence like a sink.

FINAL ANSWER: $\text{grad } f$: steepest-rise direction/rate. $\text{div } F$: local source/sink or net outward flux density.

Question 6: Angle between surfaces, scalar potential and a line integral

TOPIC

Gradient normals and conservative fields

BASIC

The angle between two surfaces is the angle between their normal vectors, obtained from gradients. A field is conservative/irrotational when $\text{curl } F = 0$ (on a suitable simply connected domain), and then $F = \text{grad } \phi$. A line integral uses $F(\mathbf{r}(\theta)) \cdot \mathbf{r}'(\theta)$.

PROCEDURE

Surfaces: write $F_1 = 0, F_2 = 0 \rightarrow$ gradients $\rightarrow$ dot-product angle. Potential: integrate one component $\rightarrow$ compare remaining components. Line integral: parameterize curve $\rightarrow$ substitute $\rightarrow$ dot with derivative $\rightarrow$ integrate limits.

6(a) - Angle between $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$

$$F_1 = x^2 + y^2 + z^2 - 9, \quad \text{grad } F_1 = (2x, 2y, 2z) = (4, -2, 4)$$

$$F_2 = x^2 + y^2 - z - 3, \quad \text{grad } F_2 = (2x, 2y, -1) = (4, -2, -1)$$

$$\cos(\theta) = |\mathbf{n}_1 \cdot \mathbf{n}_2| / (|\mathbf{n}_1| |\mathbf{n}_2|) = 16 / (6\sqrt{21}) = 8 / (3\sqrt{21})$$

FINAL ANSWER: $\theta = \cos^{-1}[8 / (3\sqrt{21})]$ approximately 54.4 degrees.

6(b) - Show the field is irrotational and find its scalar potential

$$\mathbf{F} = (y^2 + 2xz^2)\mathbf{i} + (2xy - z)\mathbf{j} + (2x^2z - y + 2z)\mathbf{k}$$

Check curl quickly:

$$R_y - Q_z = (-1) - (-1) = 0$$

$$P_z - R_x = 4xz - 4xz = 0$$

$$Q_x - P_y = 2y - 2y = 0$$

Hence $\text{curl } F = 0$.

Now integrate $\phi_x = P$:

$$\phi = xy^2 + x^2z^2 + g(y,z)$$

Compare $\phi_y = 2xy + g_y$ with $Q = 2xy - z$: $g_y = -z$, so $g = -yz + h(z)$.

Compare $\phi_z = 2x^2z - y + h'(z)$ with $R = 2x^2z - y + 2z$: $h'(z) = 2z$, so $h = z^2 + C$.

FINAL ANSWER: $\phi(x,y,z) = xy^2 + x^2z^2 - yz + z^2 + C$.

6(c) - Line integral on the spiral

$$r(\theta) = (a \cos \theta, a \sin \theta, a \theta), \quad 0 \leq \theta \leq \pi/2$$

The paper gives $F = r^2 \mathbf{i}$. Using $r^2 = x^2 + y^2 + z^2$,

$$r^2 = a^2(\cos^2 \theta + \sin^2 \theta + \theta^2) = a^2(1 + \theta^2)$$

$$dr = (-a \sin \theta, a \cos \theta, a) d\theta$$

$$F \cdot dr = a^2(1 + \theta^2)(-a \sin \theta) d\theta$$

$$\text{Integral} = -a^3 \int_0^{\pi/2} (1 + \theta^2) \sin \theta \, d\theta$$

Use integration by parts: $\int \theta^2 \sin \theta \, d\theta$ from 0 to $\pi/2$ equals $\pi/2$; $\int \sin \theta \, d\theta$ equals 1.

$$\text{Integral} = -a^3[1 + (\pi/2)] = -a^3(1 + \pi/2)$$

FINAL ANSWER: $\int_C F \cdot dr = -a^3(1 + \pi/2)$.

Question 7: Surface flux and key Laplace-transform properties

TOPIC

Flux integrals and transform identities

BASIC

For an oriented surface, flux is $\int F \cdot n \, dS$. On a unit sphere the outward unit normal is $n = (x, y, z)$. Laplace properties turn time shifts/scalings/products into simple operations on $F(s)$.

PROCEDURE

Flux: choose a surface parameterization and compute $F \cdot n \, dS$. Laplace properties: write the formula plus conditions. For $\sin(at)/t$, use the division-by- t property or integrate a parameter.

7(a) - Flux over the first-octant part of the unit sphere

$$F = yz \mathbf{i} + zx \mathbf{j} + xy \mathbf{k}, \quad x^2 + y^2 + z^2 = 1$$

On the unit sphere, outward $n = (x, y, z)$, so

$$F \cdot n = xyz + xyz + xyz = 3xyz$$

Use spherical coordinates for the first octant:

$$x = \sin \phi \cos \theta, \quad y = \sin \phi \sin \theta, \quad z = \cos \phi, \quad dS = \sin \phi \, d\phi \, d\theta$$

$$0 \leq \phi \leq \pi/2, \quad 0 \leq \theta \leq \pi/2$$

$$\text{Flux} = 3 \int_0^{\pi/2} \int_0^{\pi/2} \sin^3 \phi \cos \phi \cos(\theta) \sin(\theta) \, d\theta \, d\phi$$

$$= 3(1/4)(1/2) = 3/8$$

FINAL ANSWER: Flux = 3/8 (outward orientation).

7(b) - Laplace properties requested in the paper

Let $L\{f(t)\}=F(s)$.

- First shifting: $L\{e^{at} f(t)\}=F(s-a)$.
- Second shifting: $L\{u(t-a) f(t-a)\}=e^{-as}F(s)$.
- Scaling ($a>0$): $L\{f(at)\}=(1/a)F(s/a)$.
- Multiplication by t^n : $L\{t^n f(t)\}=(-1)^n d^n F/ds^n$.
- Division by t : $L\{f(t)/t\}=\int_s^\infty F(u) du$, when the integral exists.
- Convolution theorem: $L^{-1}\{F(s)G(s)\}=(f*g)(t)=\int_0^t f(u)g(t-u)du$.

FINAL ANSWER: The six formulas above are the requested revision set.

7(c) - $L\{\sin(at)/t\}$ and existence of $L\{\cos(at)/t\}$

Because $L\{\sin(at)\}=a/(s^2+a^2)$, the division-by- t property gives

$$L\{\sin(at)/t\}=\int_s^\infty a/(u^2+a^2) du = \tan^{-1}(a/s)$$

For $\cos(at)/t$, near $t=0$ we have $\cos(at)/t$ approximately $1/t$. Therefore the defining Laplace integral contains a logarithmically divergent integral near zero.

FINAL ANSWER: $L\{\sin(at)/t\}=\tan^{-1}(a/s)$. $L\{\cos(at)/t\}$ does not exist as an ordinary Laplace transform.

Question 8: Direct Laplace transforms, inverse transforms and convolution

TOPIC

Laplace toolkit

BASIC

Use $L\{t^n f\}=(-1)^n F^{(n)}(s)$, first shifting for e^{-at} , standard sine/cosine pairs, partial fractions, completing the square, and the logarithm identity. Convolution in time corresponds to multiplication in s .

PROCEDURE

Direct transform: identify standard $F(s)$ -> apply rule. Inverse: algebra/partial fractions -> standard pairs. Convolution: factor expression as $F(s)G(s)$, invert each, then integrate from 0 to t .

8(a) - Find the three Laplace transforms

(i) $t^2 \sin 2t$

$$L\{\sin 2t\}=2/(s^2+4)$$

$$L\{t^2 \sin 2t\}=d^2/ds^2 [2/(s^2+4)] = 4(3s^2-4)/(s^2+4)^3$$

(ii) $e^{-4t} \sin 3t/t$

$$L\{\sin 3t/t\}=\tan^{-1}(3/s)$$

$$\text{First shift } s \rightarrow s+4: L\{e^{-4t}\sin 3t/t\}=\tan^{-1}[3/(s+4)]$$

(iii) Unit step

$$L\{u(t-a)\}=e^{-as}/s. \text{ In particular, } L\{u(t)\}=1/s.$$

FINAL ANSWER: (i) $4(3s^2-4)/(s^2+4)^3$; (ii) $\tan^{-1}[3/(s+4)]$; (iii) e^{-as}/s (or $1/s$ for $a=0$).

8(b) - Find the inverse Laplace transforms

(i) $(s^2+3)/[s(s^2+9)]$

$$(s^2+3)/[s(s^2+9)] = (1/3)(1/s) + (2/3)s/(s^2+9)$$

$$f_1(t) = 1/3 + (2/3)\cos 3t$$

(ii) $s/(s^2+4s+13)$

$$s^2+4s+13 = (s+2)^2+9, \quad s = (s+2)-2$$

$$f_2(t) = e^{-2t}\cos 3t - (2/3)e^{-2t}\sin 3t$$

(iii) $\log[(s^2-1)/s^2]$

$$\log[(s^2-1)/s^2] = \log[(s-1)/s] + \log[(s+1)/s]$$

Use $L\{(e^{-at}-e^{-bt})/t\} = \log[(s+b)/(s+a)]$. Then

$$L^{-1}\{\log[(s-1)/s]\} = (1-e^{-t})/t$$

$$L^{-1}\{\log[(s+1)/s]\} = (1-e^{-(-t)})/t$$

$$f_3(t) = [2-e^{-t}-e^{-(-t)}]/t = 2(1-\cosh t)/t$$

FINAL ANSWER: (i) $1/3 + (2/3)\cos 3t$; (ii) $e^{-2t}[\cos 3t - (2/3)\sin 3t]$; (iii) $[2-e^{-t}-e^{-(-t)}]/t$.

8(c) - Use convolution theorem

$$\text{Find } L^{-1}\{s^2 / [(s^2+1)(s^2+2)]\}$$

Factor it as a product of two standard transforms:

$$s^2/[(s^2+1)(s^2+2)] = [s/(s^2+1)] [s/(s^2+2)]$$

$$L^{-1}\{s/(s^2+1)\} = \cos t, \quad L^{-1}\{s/(s^2+2)\} = \cos(\sqrt{2}t)$$

Therefore by convolution

$$f(t) = \int_0^t \cos u * \cos(\sqrt{2}(t-u)) \, du$$

Evaluating the elementary trigonometric integral gives

$$f(t) = \sqrt{2} \sin(\sqrt{2}t) - \sin t$$

FINAL ANSWER: $L^{-1}\{s^2/[(s^2+1)(s^2+2)]\} = \sqrt{2}\sin(\sqrt{2}t) - \sin t$.

Rapid Revision - procedures to memorize

Topic	Fast procedure
Linear dependence	Set $c_1v_1 + \dots = 0$; any nonzero coefficient solution means dependent.
Rank / normal form	Row reduce $\rightarrow$ count pivots; use column operations to reach I_r with zeros.
Invertibility	Square + $\det \neq 0$ (or full rank). Non-square matrices have no ordinary inverse.
Lambda system	$\det \neq 0 \rightarrow$ unique; $\det = 0 \rightarrow$ compare equations/ranks for infinite vs none.
Eigenpair	$\det(A - \lambda I) = 0$; then solve $(A - \lambda I)v = 0$.
Cayley-Hamilton	Find $p(\lambda)$; substitute A to get $p(A) = 0$.
Vector motion	$v = r'$; $a = r''$; if a given, integrate and use initial data.
Angle of surfaces	Normals are gradients; use cosine formula on the two normals.
Potential	Check $\text{curl} = 0$; integrate one component and compare

	the rest.
Line integral	Parameterize curve; integrate $F(r(t)) \cdot r'(t)$.
Surface flux	Parameterize surface; integrate $F \cdot n \, dS$ with correct orientation.
Laplace t^n rule	$L\{t^n f\} = (-1)^n \frac{d^n F}{ds^n}$.
Inverse Laplace	Simplify/partial fractions/complete square before using table.
Convolution	Factor $H(s) = F(s)G(s)$; inverse each; integrate $f(u)g(t-u)$, 0 to t .

Answers at a glance

- 1(c) Dependent: $2v_1 - v_2 + v_3 = 0$.
- 2(a) rank=2; normal form has I2 followed by zero columns/row.
- 2(b) Only $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is invertible.
- 2(c) $A^{(-1)} = \begin{bmatrix} -2 & 1/2 \\ 1 & 0 \end{bmatrix}$.
- 3(a) Unique for $\lambda \neq \pm 1$; infinite for $\lambda = \pm 1$; no no-solution value.
- 3(b) Largest eigenvalue 1, eigenvector (3,2); same dominant eigenpair for A^{10} .
- 4(b) $A^2 = \begin{bmatrix} 16 & 24 \\ 32 & 48 \end{bmatrix}$; $A^{(-2)}$ does not exist.
- 4(c) Nontrivial family $k(37, -19, -117, 67)$.
- 5(a) $|v|=10$, $|a|=18$; at $\pi/6$, $v=(0, -6, 8)$, $a=(-18, 0, 0)$.
- 5(b) $q(t) = (2t^3 + 2t + 1)i + (2t^4 + 3t + 1)j + (t - \sin t)k$.
- 6(a) $\theta = \cos^{-1}[8/(3\sqrt{21})] \sim 54.4$ degrees.
- 6(b) $\phi = xy^2 + x^2z^2 - yz + z^2 + C$.
- 6(c) $a^3(1 - \pi)$.
- 7(a) Flux = $3/8$.
- 7(c) $L\{\sin(at)/t\} = \tan^{-1}(a/s)$; ordinary $L\{\cos(at)/t\}$ does not exist.
- 8(a) $4(3s^2 - 4)/(s^2 + 4)^3$; $\tan^{-1}[3/(s+4)]$; $e^{(-as)}/s$.
- 8(b) $1/3 + (2/3)\cos 3t$; $e^{(-2t)}[\cos 3t - (2/3)\sin 3t]$; $[2 - e^{-t} - e^{(-t)}]/t$.
- 8(c) $\sqrt{2}\sin(\sqrt{2}t) - \sin t$.

Beginner exam strategy

- Attempt definition/property questions first to secure easy marks.
- In a matrix problem, check dimensions and determinant/rank before trying an inverse.
- For vector calculus, write component names P, Q, R before curl/divergence calculations.
- For surface-angle questions, remember: angle of surfaces = angle of their normals.
- For Laplace questions, first rewrite into a standard transform form; do algebra before inversion.
- Write row operations and intermediate steps. Even with an arithmetic slip, the method remains visible.